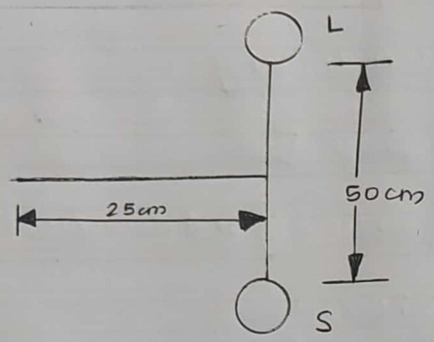
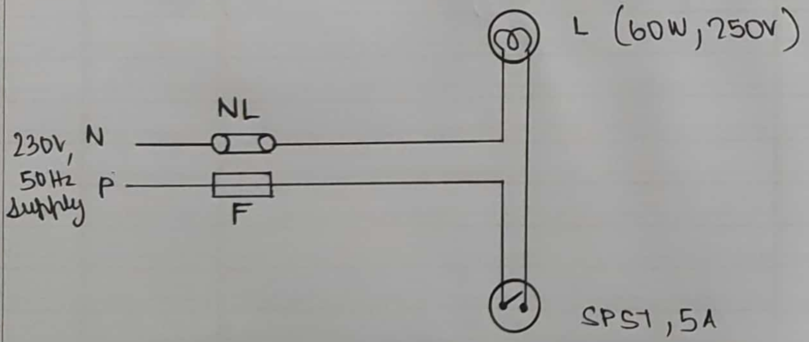
LAYOUT DIAGRAM



CIRCUIT DIAGRAM



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ONE LAMP ONE SWITCH

AIM

Wire up a circuit to obtain one lamp controlled by one switch and prepare an estimation for the same.

DESIGN

Power, $P = VI\cos\phi$

$$I = \frac{P}{V}$$

$P = 60W$

$V = 230V$

$$I = \frac{60}{230} = \underline{\underline{0.26A}}$$

Fuse wire rating,

$= \text{load current} \times 1.5A$

$= 0.26 \times 1.5$

$= \underline{\underline{0.39A}}$

OBSERVATION TABLE

SWITCH	LAMP
OFF	DARK
ON	BRIGHT

NOTE:

L - Lamp
S - Switch
NL - Neutral Link
F = Fuse
SPST - Single Pole Single Throw
N - Neutral
P - Phase

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MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1)	Switch	SPST (single pole single throw)	5A 250V	1
2)	Lamp	Incandescent	60W 250V	1
3)	Conduit	20mm PVC		75cm
4)	Copper Wire	1 sq mm		175 cm
5)	lamp holder	straight batten	5 A	1
6)	Junction	20mm 3way PVC		1
7)	Round Block	3inch PVC		1
8)	Switch Box	3x2" PVC		1
9)	Clamp / Saddle	20mm metal		3
10)	Fuse Unit	Kit Kat (ordinary rewirable	16 A	1

11)	Fuse wire	Tin & lead alloy	2 A	10cm
12)	Neutral link	Porcalene base with copper segment	16 A	1
13)	Insulation Tape			1
14)	Screw	1/2 "		6
		3/4 "		4
		1 1/4 "		1

TOOLS REQUIRED

Screw driver, line tester, combination player, ball-peen hammer, poker, nose player, wire stripper.

PROCEDURE

1. Draw the circuit diagram and check whether it is correct.
2. Fix the conduit to the board using saddle of appropriate diameter
3. Wire the circuit as per the circuit diagram

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PROCEDURE

1. Draw the circuit diagram and check whether it is correct.
2. Fix the conduit to the board using saddle of appropriate diameter
3. Wire the circuit as per the circuit diagram

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4. Fix the switch box and give its connection

5. Fix the round block to the base board

6. Fix the lamp holder on the top of the round block and give its connection.

7. Attach the neutral link and fuse.

8. Add the incandescent lamp to the holder and turn the switch on.

9. Record observation on the observation table

RESULT

Condition have been satisfied. The circuit was wired up and an estimation was created for the same.

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29/11